Expert Systems with Applications

Highlights

Agentic SABRE: An Uncertainty-Aware Neuro-Symbolic Multi-Agent Framework for Adaptive Ransomware Detection

- Neuro-symbolic multi-agent SABRE fuses static and behavioural ransomware evidence
- Monte Carlo Dropout provides calibrated epistemic uncertainty for triage decisions
- Two-threshold policy automates containment or escalates uncertain borderline cases
- CTGAN score-space augmentation improves fusion robustness under class imbalance
- Gradient saliency and counterfactual XAI explain agent and orchestrator decisions